EXPERIMENT - 4

AIM: Using the Employee schema, write queries with INNER JOIN, LEFT JOIN, selfjoin, 3-way join, correlated subqueries, EXISTS, and simulated INTERSECT and EXCEPT. Compare execution plans using EXPLAIN.

SOURCE CODE:

```
CREATE TABLE Department ( dept_id
   INT PRIMARY KEY,
   dept_name VARCHAR(50)
);

CREATE TABLE Project (
   project_id INT PRIMARY KEY,
   project_name VARCHAR(50),
   dept_id INT,
   FOREIGN KEY (dept_id) REFERENCES Department(dept_id)
);

CREATE TABLE Employee ( emp_id
   INT PRIMARY KEY,
   emp_name VARCHAR(50),
   dept_id  INT,  project_id
   INT,                salary
   DECIMAL(10,2),    city
   VARCHAR(50),
   FOREIGN KEY (dept_id) REFERENCES Department(dept_id),
   FOREIGN KEY (project_id) REFERENCES Project(project_id)
);

INSERT INTO Department VALUES
(1, 'HR'), (2, 'Finance'), (3, 'IT'), (4, 'Marketing'), (5, 'Operations');

INSERT INTO Project VALUES
(101, 'Recruitment System', 1), (102, 'Payroll Automation', 2),
(103, 'Cybersecurity Upgrade', 3), (104, 'Website Revamp', 3),
(105, 'Ad Campaign', 4), (106, 'Market Research', 4),
(107, 'Logistics Optimization', 5), (108, 'Inventory System', 5);

INSERT INTO Employee VALUES
(1, 'Alice', 1, 101, 45000, 'Delhi'),
(2, 'Bob', 1, 101, 42000, 'Mumbai'),
(3, 'Charlie', 1, 101, 47000, 'Pune'),
(4, 'David', 2, 102, 55000, 'Delhi'),
(5, 'Eva', 2, 102, 60000, 'Lucknow'),
(6, 'Frank', 2, 102, 52000, 'Chennai'),
(7, 'Grace', 3, 103, 70000, 'Bangalore'),
(8, 'Hannah', 3, 103, 68000, 'Hyderabad'),
(9, 'Ian', 3, 104, 72000, 'Delhi'),
(10, 'Jack', 3, 104, 65000, 'Pune'),
(11, 'Karen', 3, 104, 64000, 'Mumbai'),
```

```
(12, 'Leo', 4, 105, 48000, 'Delhi'),
(13, 'Mia', 4, 105, 50000, 'Lucknow'),
(14, 'Nina', 4, 106, 51000, 'Chennai'),
(15, 'Oscar', 4, 106, 53000, 'Delhi'),
(16, 'Paul', 5, 107, 56000, 'Hyderabad'),
(17, 'Quinn', 5, 107, 57000, 'Bangalore'),
(18, 'Rita', 5, 107, 59000, 'Delhi'),
(19, 'Sam', 5, 108, 60000, 'Lucknow'),
(20, 'Tina', 5, 108, 61000, 'Mumbai'),
(21, 'Uma', 5, 108, 62000, 'Chennai'),
(22, 'Victor', 2, 102, 53000, 'Delhi'),
(23, 'Wendy', 2, 102, 54000, 'Hyderabad'),
(24, 'Xavier', 3, 103, 71000, 'Bangalore'),
(25, 'Yara', 3, 104, 66000, 'Lucknow'),
(26, 'Zane', 4, 105, 49000, 'Delhi'),
(27, 'Aditi', 4, 106, 52000, 'Mumbai'),
(28, 'Bhavya', 1, 101, 46000, 'Chennai'),
(29, 'Chetan', 1, 101, 43000, 'Lucknow'), (30,
'Devika', 5, 107, 58000, 'Delhi');

SELECT E.emp_id, E.emp_name, D.dept_name
FROM Employee E
INNER JOIN Department D
   ON E.dept_id = D.dept_id;

SELECT D.dept_id, D.dept_name, E.emp_name
FROM Department D
LEFT JOIN Employee E
   ON D.dept_id = E.dept_id;

SELECT E1.emp_name AS Employee1, E2.emp_name AS Employee2, E1.dept_id
FROM Employee E1
INNER JOIN Employee E2
   ON E1.dept_id = E2.dept_id
   AND E1.emp_id < E2.emp_id;

SELECT E.emp_name, P.project_name, D.dept_name
FROM Employee E
INNER JOIN Project P
   ON E.project_id = P.project_id
INNER JOIN Department D
   ON E.dept_id = D.dept_id;

SELECT E.emp_name, E.dept_id, E.salary
FROM Employee E
WHERE    E.salary    >    (
   SELECT AVG(E2.salary)
   FROM Employee E2
   WHERE E2.dept_id = E.dept_id );
SELECT E.emp_id, E.emp_name, E.dept_id
```

```
FROM Employee E
WHERE EXISTS ( SELECT
   1
   FROM Project P
   WHERE P.dept_id = E.dept_id );

SELECT emp_name
FROM Employee
WHERE emp_id IN (
   SELECT emp_id
   FROM Employee
   WHERE city = 'Delhi' )
AND emp_id IN (
   SELECT emp_id
   FROM Employee
   WHERE salary > 60000 );

SELECT emp_name
FROM Employee
WHERE emp_id NOT IN (
   SELECT emp_id
   FROM Employee
   WHERE city = 'Delhi' );

EXPLAIN
SELECT E.emp_name, D.dept_name
FROM Employee E
INNER JOIN Department D
   ON E.dept_id = D.dept_id;

EXPLAIN
SELECT D.dept_name, E.emp_name
FROM Department D
LEFT JOIN Employee E
   ON D.dept_id = E.dept_id;

EXPLAIN
SELECT E.emp_name, E.salary
FROM Employee E
WHERE   E.salary   >   (
   SELECT AVG(E2.salary)
   FROM Employee E2
   WHERE E2.dept_id = E.dept_id );

EXPLAIN
SELECT E.emp_name FROM Employee E
WHERE EXISTS ( SELECT
   1
   FROM Project P
   WHERE P.dept_id = E.dept_id );
```

OUTPUT :-

AMARJEET SRIVASTAVA

I/O | AI Agent | 91 ms

STDIN

Input for the program (Optional)

Output:

```
| emp_id | emp_name | dept_id |
+--------+----------+---------+
|      1 | Alice    |       1 |
|      2 | Bob      |       1 |
|      3 | Charlie  |       1 |
|      4 | David    |       2 |
|      5 | Eva      |       2 |
|      6 | Frank    |       2 |
|      7 | Grace    |       3 |
|      8 | Hannah   |       3 |
|      9 | Ian      |       3 |
|     10 | Jack     |       3 |
|     11 | Karen    |       3 |
|     12 | Leo      |       4 |
|     13 | Mia      |       4 |
|     14 | Nina     |       4 |
|     15 | Oscar    |       4 |
|     16 | Paul     |       5 |
|     17 | Quinn    |       5 |
|     18 | Rita     |       5 |
|     19 | Sam      |       5 |
|     20 | Tina     |       5 |
|     21 | Uma      |       5 |
|     22 | Victor   |       2 |
|     23 | Wendy    |       2 |
|     24 | Xavier   |       3 |
|     25 | Yara     |       3 |
|     26 | Zane     |       4 |
|     27 | Aditi    |       4 |
```

AMARJEET SRIVASTAVA

I/O | AI Agent | 91 ms

STDIN

Input for the program (Optional)

Output:

```
|     28 | Bhavya   |       1 |
|     29 | Chetan   |       1 |
|     30 | Devika   |       5 |
+--------+----------+---------+
+----------+
| emp_name |
+----------+
| Ian      |
+----------+
+----------+
| emp_name |
+----------+
| Bob      |
| Charlie  |
| Eva      |
| Frank    |
| Grace    |
| Hannah   |
| Jack     |
| Karen    |
| Mia      |
| Nina     |
| Paul     |
| Quinn    |
| Sam      |
| Tina     |
| Uma      |
| Wendy    |
| Xavier   |
```

AMARJEET SRIVASTAVA

I/O | AI Agent | 91 ms

STDIN

Input for the program (Optional)

Output:

```
| Xavier   |
| Yara     |
| Aditi    |
| Bhavya   |
| Chetan   |
+----------+
+----+-------------+-------+------------+-------+---------------+---------+---------+-------------------------------+------+----------+-------------+
| id | select_type | table | partitions | type  | possible_keys | key     | key_len | ref                           | rows | filtered | Extra       |
+----+-------------+-------+------------+-------+---------------+---------+---------+-------------------------------+------+----------+-------------+
|  1 | SIMPLE      | E     | NULL       | ALL   | dept_id       | NULL    | NULL    | NULL                          |   30 |   100.00 | Using where |
|  1 | SIMPLE      | D     | NULL       | eq_ref | PRIMARY      | PRIMARY | 4       | sandbox_db.E.dept_id |    1 |   100.00 | NULL        |
+----+-------------+-------+------------+-------+---------------+---------+---------+-------------------------------+------+----------+-------------+
+----+-------------+-------+------------+------+---------------+------+---------+------+------+----------+-----------------------------------------------------+
| id | select_type | table | partitions | type | possible_keys | key  | key_len | ref  | rows | filtered | Extra                                               |
+----+-------------+-------+------------+------+---------------+------+---------+------+------+----------+-----------------------------------------------------+
|  1 | SIMPLE      | D     | NULL       | ALL  | NULL          | NULL | NULL    | NULL |    5 |   100.00 | NULL                                                |
|  1 | SIMPLE      | E     | NULL       | ALL  | dept_id       | NULL | NULL    | NULL |   30 |   100.00 | Using where; Using join buffer (hash join) |
+----+-------------+-------+------------+------+---------------+------+---------+------+------+----------+-----------------------------------------------------+
+----+--------------------+-------+------------+------+---------------+------+---------+------+------+----------+-------------+
| id | select_type        | table | partitions | type | possible_keys | key  | key_len | ref  | rows | filtered | Extra       |
+----+--------------------+-------+------------+------+---------------+------+---------+------+------+----------+-------------+
|  1 | PRIMARY            | E     | NULL       | ALL  | NULL          | NULL | NULL    | NULL |   30 |   100.00 | Using where |
|  2 | DEPENDENT SUBQUERY | E2    | NULL       | ALL  | dept_id       | NULL | NULL    | NULL |   30 |    10.00 | Using where |
+----+--------------------+-------+------------+------+---------------+------+---------+------+------+----------+-------------+
+----+-------------+--------------+------------+--------+-----------------+-----------------+---------+-------------------------------+------+----------+-------------+
| id | select_type | table        | partitions | type   | possible_keys   | key             | key_len | ref                           | rows | filtered | Extra       |
+----+-------------+--------------+------------+--------+-----------------+-----------------+---------+-------------------------------+------+----------+-------------+
|  1 | SIMPLE      | E            | NULL       | ALL    | dept_id         | NULL            | NULL    | NULL                          |   30 |   100.00 | Using where |
|  1 | SIMPLE      | <subquery2>  | NULL       | eq_ref | <auto_distinct_key> | <auto_distinct_key> | 5       | sandbox_db.E.dept_id |    1 |   100.00 | NULL        |
|  2 | MATERIALIZED | P           | NULL       | index  | dept_id         | dept_id         | 5       | NULL                          |    8 |   100.00 | Using index |
```